

Ans 1. The SI unit of force is the Newton (N), defined as the force needed to accelerate a 1 kg mass at 1 m/s^2 .

Ans 2. Newton's second law states that the force acting on an object equals the rate of change of its momentum, $F = ma$.

Ans 3. Acceleration is the rate of change of velocity with time. Its SI unit is metre per second squared (m/s^2).

Ans 4. Kinetic energy $KE = \frac{1}{2} m v^2$, where m is mass and v is velocity.

Ans 5. A force can change the speed of an object, and it can change the direction of motion of an object.